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Source: *Journal of the American Statistical Association*, Vol. 75, No. 372 (Dec., 1980), pp. 1010-1014
Published by: Taylor & Francis, Ltd. on behalf of the American Statistical Association
Stable URL: https://www.jstor.org/stable/2287197
Accessed: 12-01-2026 20:33 UTC

Test of Trend in Count Data: Multinomial Distribution Case

YOUNG JACK LEE*

Given k multinomial probabilities $p_i(\sum p_i = 1)$, let p_i be a known continuous function $f(w_i, \pi_i)$ of a known weight constant w_i and an unknown constant π_i. The function f is assumed strictly monotone in π. The maximin statistic for testing the homogeneity of the π_i's versus a trend in the π_i's is derived. The Armitage statistic for testing linear trend is maximin for a certain alternative hypothesis space. The methods are applied to lying habit data (Maxwell 1961), virology data (Gart 1964), and multiple primary neoplasms data (Bellet et al. 1977).

KEY WORDS: Test of trend; Maximin method; Multinomial distribution; Poisson distribution.

1. INTRODUCTION

We consider multinomial probabilities $p_1, \ldots, p_k$, which are indexed by an ordinal attribute that defines the categories $1, 2, \ldots, k$. Under the null hypothesis $p_1 = p_2 = \ldots = p_k$, the attribute has no influence on cell probabilities. In this article, we present maximin statistics against quite general classes of ordered alternatives. Using the method of Lee (1977), one can test the null hypothesis H: $p_1 = p_2 = \ldots = p_k = 1/k$ against alternatives such as K: $p_{i+1} \geq \lambda_i^* p_i$, $\lambda_i^* \geq 1$, $i = 1, 2, \ldots, k-1$. We generalize the method to $p_i = f(w_i, \pi_i)$ where f is a positive function that is strictly increasing in π_i. We consider the null hypothesis $H: p_i = f(w_i, \pi_0)$, with π_0 defined implicitly by $\sum p_i = 1$, versus the alternative $K: p_i = f(w_i, \pi_i)$, with $\sum p_i = 1$ and $\pi_{i+1} \geq \lambda_i^* \pi_i$, $\lambda_i^* \geq 1$, $i = 1, 2, \ldots, k-1$.

The method presented in this article is the maximin statistic. The maximin statistic maximizes the minimum power over an alternative hypothesis space separated from the null hypothesis by an indifference zone (Lee 1977). It can detect the possible trend without specifying a functional relationship and requires a very simple computation. The maximin method may be useful for preliminary data analysis preceding more detailed analysis in the case of significant trend, and may satisfy the needs of experimenters who are only willing to postulate a possible trend. The method is exact regardless of the sample size, and it will be shown in Section 3 that the kernel of χ^2_0 statistic of Armitage (1955) is the maximin statistic for a certain alternative hypothesis space.

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2. NOTATIONS AND DEFINITIONS

Let the probability of the ith event be a known continuous function $f(w_i, \pi_i)$ of a known weight constant w_i and an unknown nonnegative constant π_i. The function $f(w_i, \pi_i)$ is assumed to be strictly increasing in π_i. The random variable associated with the ith event is denoted by Y_i and its observed count by y_i. The vector of multinomial observation is denoted by $\tilde{y} = (y_1, y_2, \ldots, y_k)$, $\sum y_i = n$, $y_i \geq 0$.

The space of ordered π_i's is denoted by

$$\Omega = \{\tilde{\pi}: \tilde{\pi} = (\pi_1, \ldots, \pi_k),\ 0 \leq \pi_1 \leq \pi_2 \leq \ldots \leq \pi_k,\ \sum f(w_i, \pi_i) = 1\} .$$

A subspace of Ω is defined for a $(k-1)$-tuple $\tilde{\lambda}^* = (\lambda_1^*, \ldots, \lambda_{k-1}^*)$ of constants not smaller than one (and at least one strictly greater than one). The subspace is denoted by

$$\Omega_k(\tilde{\lambda}^*) = \{\tilde{\pi}: \tilde{\pi} \in \Omega,\ \pi_{i+1} \geq \lambda_i^* \pi_i,\ 1 \leq i \leq k-1\} .$$

The singleton subspace of Ω with homogeneous π_i's is denoted by $\{\tilde{\pi}(H)\} = \{(\pi_0, \ldots, \pi_0)\}$ where π_0 satisfies $\sum f(w_i, \pi_0) = 1$ and A denotes the collection of $\tilde{a} = (a_1, a_2, \ldots, a_k) \in R^k$ satisfying $-\infty < a_1 \leq a_2 \leq \ldots \leq a_k < \infty$ with at least one strict inequality.

The hypothesis testing problem considered is

$$\langle H: \tilde{\pi} = \tilde{\pi}(H) \text{ vs. } K: \tilde{\pi} \in \Omega_k(\tilde{\lambda}^*)\rangle, \quad \text{denoted by } \langle H, K | \tilde{\lambda}^*\rangle .$$

We assume that $\Omega_k(\tilde{\lambda}^*)$ is not a singleton, because otherwise the most powerful test exists for $\langle H, K | \tilde{\lambda}^*\rangle$.

The maximin for $\langle H, K | \tilde{\lambda}^*\rangle$ is the test that maximizes minimum power over the space $\Omega_k(\tilde{\lambda}^*)$. In the case of $\langle H: \tilde{\pi} = \pi(H)$ vs. $K: \pi \in \Omega - \{\tilde{\pi}(H)\}\rangle$, power would be the size of the test and would occur at $\tilde{\pi}(H)$, and the maximin test is thus not considered.

Whenever a summation index runs from 1 to k, for notational convenience the range of the index will be omitted for the summation sign.

© Journal of the American Statistical Association
December 1980, Volume 75, Number 372
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3. MAXIMIN TESTS

Consider a size α test of the following form:

$$\begin{aligned}\psi_\alpha(\tilde{y}) &= 1 \text{ if } V(\tilde{a}, \tilde{y}) > v_\alpha(\tilde{a}) \\ &= \gamma \text{ if } V(\tilde{a}, \tilde{y}) = v_\alpha(\tilde{a}) \\ &= 0 \text{ if } V(\tilde{a}, \tilde{y}) < v_\alpha(\tilde{a}) \ ,\end{aligned}$$

where $\tilde{a} \in A$, $V(\tilde{a}, \tilde{y}) = \sum a_i y_i$, and $v_\alpha(\tilde{a})$ and γ[1] $(0 \leq \gamma < 1)$ are chosen to satisfy significance level $E\{\psi_\alpha(\tilde{Y})|H\} = \alpha$. Denote this class of size α tests by $\Psi_\alpha(A)$.

Theorem 1: Suppose $f(w_i, \pi_i)$ and $\tilde{\lambda}^*$ are specified. Then the size α maximin test for $\langle H, K|\tilde{\lambda}^*\rangle$ exists and is $\psi_{a^*}(\tilde{y})$ in $\Psi_\alpha(A)$ with

$$a_i^* = \ln \{f(w_i, \pi_i^*)/f(w_i, \pi_0)\}$$

and

$$\tilde{\pi}^* = (\pi, \pi\lambda_1^*, \pi\lambda_1^*\lambda_2^*, \ldots, \pi\lambda_1^*\lambda_2^* \ldots \lambda_{k-1}^*) \ .$$

The π is determined to satisfy the constraint $\sum f(w_i, \pi_i^*) = 1$.

Proof: Suppose that $\tilde{\pi} = \tilde{\pi}(H)$ under H and $\tilde{\pi} = \tilde{\pi}' \in \Omega_k(\tilde{\lambda}^*)$ under K where $\tilde{\pi}'$ is arbitrarily fixed. The most powerful size α test for $\tilde{\pi}(H)$ versus $\tilde{\pi}'$ can be shown to be in $\Psi_\alpha(A)$. Denote the test by $\psi_{a'}(\tilde{y})$. Since the $\Omega_k(\tilde{\lambda}^*)$ is a closed set and the power function is continuous, there exists a $\tilde{\pi}$ in $\Omega_k(\tilde{\lambda}^*)$ at which the power of the test $\psi_{a'}(\tilde{y})$ is minimized. In fact Lemma 3 (in Appendix) shows that for each $\psi_a(\tilde{y}) \in \Psi_\alpha(A)$,

$$\min_{\tilde{\pi} \in \Omega_k(\tilde{\lambda}^*)} E\{\psi_a(\tilde{Y})|\tilde{\pi}\} = E\{\psi_a(\tilde{Y})|\tilde{\pi}^*\} \ .$$

The most powerful test $\psi_{a^*}(\tilde{y})$ for testing $\tilde{\pi}(H)$ against $\tilde{\pi}^*$ is in $\Psi_\alpha(A)$ and satisfies, for each $\psi_a(\tilde{y}) \in \Psi_\alpha(A)$,

$$E\{\psi_{a^*}(\tilde{Y})|\tilde{\pi}^*\} \geq E\{\psi_a(\tilde{Y})|\tilde{\pi}^*\} \ .$$

This demonstrates that $\psi_{a^*}(\tilde{y})$ is the maximin test for $\langle H, K|\tilde{\lambda}^*\rangle$ and the associated statistic is

$$V(\tilde{a}^*, \tilde{y}) = \sum y_i \log \{f(w_i, \pi_i^*)/f(w_i, \pi_0)\} \ . \quad (3.1)$$

The maximin statistic can be derived from (3.1) if $\tilde{\lambda}^*$ is specified. We now consider three important special cases of maximin statistics for which the specification of $\tilde{\lambda}^*$ is not required. For all these special cases and for the rest of this section, assume that $f(w_i, \pi_i) = w_i\pi_i$.

Corollary 1: Suppose that an unspecified $\tilde{\lambda}^*$ satisfies $\lambda_1^* = \ldots = \lambda_{k-1}^* > 1$, namely, a strict trend alternative. Then the maximin statistic for $\langle H, K|\tilde{\lambda}^*\rangle$ is

$$\sum iy_i \ . \quad (3.2)$$

[1] The randomization constant γ is to ensure the test has the given significance level. In practice one chooses the critical region so that the size of the test may be as close to the given nominal significance level as possible without randomization. In the case of a large sample, the randomization constant becomes negligible. The γ is not required for computing "p values," the least significance level to reject H. For the theoretical development, however, we keep the γ.

Corollary 2: Suppose that $x_1, \ldots, x_k$ are monotonically increasing influencing variables for the k categories $E_1, \ldots, E_k$ and that $\lambda_i^* = \exp(x_{i+1} - x_i)$, $i = 1, 2, \ldots, k-1$. Then the maximin test statistic for $\langle H, K|\tilde{\lambda}^*\rangle$ is

$$\sum x_i y_i \ . \quad (3.3)$$

Corollary 3: If $\lambda_{k-t}^* > 1$ and $\lambda_i^* = 1$ for $i \neq k-t$, namely, $\pi_1 \leq \pi_2 \leq \ldots \leq \pi_{k-t} < \pi_{k-t+1} \leq \pi_k$, then the maximin statistic for $\langle H, K|\tilde{\lambda}^*\rangle$ is

$$\sum_{i>k-t} y_i \ . \quad (3.4)$$

We will prove Corollary 2. Replacing each π_i^* with $\pi(\Pi_{j=1}^{i-1} \lambda_j^*)$ $(= \pi \exp(x_i - x_1))$ and $f(w_i, \pi_i)$ with $w_i\pi_i$ in (3.1), we obtain

$$\begin{aligned}V(\tilde{a}^*, \tilde{y}) &= n \log \pi/\pi_0 + \sum_{i>1} (x_i - x_1)y_i \\ &= n(\log \pi/\pi_0 - x_1) + \sum x_i y_i \ .\end{aligned}$$

The kernel of the statistic is $\sum x_i y_i$. Other proofs are similar.

The Armitage (1955) statistic χ^2_0 for linear trend in the $2 \times k$ table is to test H: $b = 0$ vs. K: $b \neq 0$ in $\pi_i = a + bx_i$ where x_i is known quantitative variable for the ith category: It can be applied to testing trends in frequencies and its kernel is $\sum x_i y_i$ (Eqs. (7) and (8) of Armitage). Corollary 2 shows that the same statistic can be derived from the maximin considerations, implying that the Armitage statistic is maximin for particular alternative space.

Unless the power calculation is required, Corollaries 1 and 3 can be applied without specifying the size of λ_i^*'s. The statistic $\sum iy_i$ is much more useful and widely applicable than Corollary 1 shows. The demonstration follows.

Theorem 2: Suppose that $\tilde{\lambda}^*$ satisfies $\min(\lambda_1^*, \ldots, \lambda_{k-1}^*) = \lambda_{\min}^* > 1$. Then the power of the test based on $\sum iy_i$ is a lower bound, in maximizing the minimum power within $\Omega_k(\tilde{\lambda}^*)$, for the power of the maximin test for $\langle H, K|\tilde{\lambda}^*\rangle$.

Proof: Replace each λ_i^* with $\lambda_{\min}^*$ and denote the resulting indifference-zone vector by $\tilde{\lambda}_{\min}^*$. Since $\Omega_k(\tilde{\lambda}^*)$ is a subset of $\Omega_k(\tilde{\lambda}_{\min}^*)$, for each $\psi_a(\tilde{y}) \in \Psi_\alpha(A)$

$$\min_{\tilde{\pi} \in \Omega_k(\tilde{\lambda}^*)} E\{\psi_a(\tilde{Y})|\tilde{\lambda}\} \geq \min_{\tilde{\pi} \in \Omega_k(\tilde{\lambda}_{\min}^*)} E\{\psi_a(\tilde{Y})|\tilde{\lambda}\} \ . \quad (3.5)$$

Thus (3.5) guarantees that the maximin test for $\langle H, K|\tilde{\lambda}_{\min}^*\rangle$ cannot be more powerful in maximizing the minimum power within $\Omega_k(\tilde{\lambda}^*)$ than that for $\langle H, K|\tilde{\lambda}^*\rangle$, and the maximin statistic for $\langle H, K|\tilde{\lambda}^*\rangle$ is $\sum iy_i$.

Theorem 2 demonstrates that the strict trend can be tested by the statistic $\sum iy_i$,

1. Without making the specific assumption on the functional relationship between influencing and dependent variables and

2. Without requiring each category to be assigned a quantitative value.

Armitage (1955) recommended the statistic based on $\sum iy_i$ for testing trend when no functional relationship can be assumed to be known (p. 378) or each category can't be quantified but is a well-defined ordered group (p. 383). Theorem 2 gives theoretical justification for his recommendation.

The statistic $\sum iy_i$ is maximin for the alternative space defined as in Corollary 1 and is conservative for the ordered alternative space with $\tilde{\lambda}_{\min}^* > 1$ because of (3.5). In this sense we may call it the conservative maximin statistic for testing for strict trend. The exact maximin test for strict trend is not usually derivable, because $\tilde{\lambda}^*$ is not specified or assumed to be known.

4. EXAMPLES

Example 1: A sample of 223 boys attending a clinic were classified by age and lying habit, as shown in Table 1 taken from Maxwell (1961, p. 64). One of the concerns of the research was to determine if there was an age trend among inveterate liars. Let Y_{ij} and θ_{ij} denote the frequency and probability of the (i, j)th cell ($i = 1, 2$, $j = 1, \ldots, 5$). Then $(Y_{11}, Y_{12}, \ldots, Y_{25})$ forms a multinomial random vector with $(223, 10, \theta_{11}, \theta_{12}, \ldots, \theta_{25})$. The question is whether there is any age trend in lying habit. To answer this question, we use the method of this article. First we note that $(Y_{11}, Y_{12}, \ldots, Y_{15})$ is a conditional random vector with

$$(95, 5, \theta_{11}/\sum_j \theta_{1j}, \ldots, \theta_{15}/\sum_j \theta_{1j}) .$$

(For convenience of discussion, denote the parameter vector by $(n, k, p_1, \ldots, p_k)$, $\sum p_i = 1$.) The preceding question on the trend is equivalent to asking whether there is an increasing trend in $(p_1, \ldots, p_k)$ after adjustment on the age-group size difference.

Suppose that w_i ($\sum w_i = 1$) is the proportion of the ith age group. Let δ_i be the conditional probability of lying given membership in age category i. Then $\theta_{1i}/w_i\delta_i$ and $\theta_{2i} = w_i(1 - \delta_i)$. Using these notations, we may write $p_i = w_i\delta_i/\sum w_j\delta_j$. If there is the age trend in lying habit, then we would have

$$\delta_1 \leq \delta_2 \leq \ldots \leq \delta_k .$$

Now denoting $\delta_i/\sum_j w_j\delta_j$ by π_i, we may write

$$p_i = w_i\pi_i (1 \leq i \leq k) .$$

If there is the age trend, then

$$\pi_1 \leq \pi_2 \leq \ldots \leq \pi_k .$$

Suppose that under the alternative the age trend is strict, namely, $\pi_{i+1}/\pi_i \geq \lambda_i^*$, $\lambda_{\min}^* > 1$ but with unknown λ_i^*'s. The statistic to test for trend is

$$V(\tilde{a}^*, \tilde{y}) = \sum iy_i .$$

Under the null hypothesis H,

$$E\{\sum iY_i | H\} = n \sum iw_i$$

and

$$\operatorname{var}\{\sum iY_i | H\} = n\{\sum i^2 w_i - (\sum iw_i)^2\} .$$

The proportion w_i of the ith age group may be estimated from the sample by $w_i = (y_{1i} + y_{2i})/223 = c_i/223$. Replacing w_i with $c_i/223$ in the mean and variance, we obtain

$$z = \{V(\tilde{a}^*, \tilde{y}) - E(\sum iY_i | H)\}/\sqrt{\operatorname{var}\{\sum iY_i | H\}} = 1.885 .$$

The asymptotic p value is .03.

Suppose that a psychiatrist hypothesizes that the trend is not strict but that the trend is marked between the elementary school age and the secondary school age, that is, under K

$$\pi_1 \leq \pi_2 \leq \pi_3 < \pi_4 \leq \pi_5, \ \pi_4/\pi_3 \geq \lambda_3^* > 1 .$$

From Corollary 3, the maximin statistic is

$$V(\tilde{a}^*, \tilde{y}) = y_4 + y_5 ,$$

the null distribution of which is binomial $B(n, w_4 + w_5)$. The asymptotic p value from the data is .04.

Example 2: The number of pock-producing viruses were considered Poisson variates and were counted at several dilutions through in vivo experiments (Gart 1964). Since the Poisson random variables form a multinomial vector conditional on the total number of observed counts, the possible trend produced by the dilutions can be tested by the maximin test. The following data is taken from Table 1 in Gart (1964).

2. Pock Count Data

Dilution No. (i) / Dilution Factor	1 / 1/8	2 / 1/4	3 / 1/2	4 / 1/1	Total
Number of eggs (r_i)	15	12	13	10	50
Number of pock counts (y_i)	21	28	79	121	249

1. Lying Habit Data

Lying Habit	Age Group 5–7 (1)	8–9 (2)	10–11 (3)	12–13 (4)	14–15 (5)	Total
Yes (1)	6	18	19	27	25	95
No (2)	15	31	31	32	19	128
Column Sum C_i	21	49	50	59	44	223

Let μ_i denote the expected number of pock count in an egg at the ith dilution. Then we have

$$E(Y_i) = r_i\mu_i \ ,$$

where r_i is the number of eggs for the ith dilution. The problem is to test

$$H: \mu_1 = \mu_2 = \ldots = \mu_k \quad \text{vs. } K: \mu_1 \leq \mu_2 \leq \ldots \leq \mu_k \ . \quad (4.1)$$

The number of pock counts form a conditional multinomial random vector with probabilities

$$(p_1, \ldots, p_k) = (r_1\mu_1/\sum r_i\mu_i, \ldots, r_k\mu_k/\sum r_i\mu_i) \ . \quad (4.2)$$

From (4.2), we note the r_j corresponds to w_j (weight constants) and $\mu_j/\sum r_i\mu_i$ to π_j and that the testing problem (4.1) is equivalent to

$$H: \pi_1 = \pi_2 = \ldots = \pi_k \quad \text{vs. } K: \pi_1 \leq \pi_2 \leq \ldots \leq \pi_k \ . \quad (4.3)$$

Suppose that K of (4.1) restricts $\mu_{i+1}/\mu_i \geq \lambda_i^* > 1$, $\lambda_{\min}^* > 1$. Then equivalently, $\pi_{i+1}/\pi_i \geq \lambda_i^* > 1$ for all i under K of (4.3). Since λ_i^*'s are not known, the statistic to use is

$$V(\tilde{a}^*, \tilde{y}) = \sum iy_i \ .$$

Since under H

$$(p_1, p_2, \ldots, p_k) = (r_1/\sum r_i, r_2/\sum r_i, \ldots, r_k/\sum r_i) \ ,$$

the asymptotic p value can be obtained. The data yields $z = 12.02$ and p value essentially zero.

Example 3: Bellet et al. (1977) cross-classified 121 Caucasian female patients with primary cutaneous malignant melanoma by years at risk and involvement of nonmelanocytic, noncutaneous malignant neoplasms, yielding the following data taken from their Table 2.

3. Multiple Neoplasms Data

Group No.	Years at Risk	Number of Subjects	The Number of Patients with Other Primary Cancers: Expected (E_i)[a]	Observed (Y_i)
1	15–24[b]	8	.02951	1
2	25–29	6	.03774	0
3	30–34	6	.0618	2
4	35–39	9	.153	0
5	40–44	11	.3135	1
6	45–49	9	.4239	0
7	50–54	10	.707	0
8	55–59	15	1.50	1
9	60–64	18	2.412	2
10	65–69	8	1.392	0
11	70–74	8	1.752	1
12	75–79	5	1.350	1
13	80+	8	2.584	2

[a] Obtained by multiplying the number of subjects in the age group with the probability that a healthy subject develops cancer during the time at risk. The probability is age specific and is estimated from the incidence rates for all nonskin cancers extracted from the Third National Cancer Survey. For details see Bellet et al. (1977).

[b] Pooled the two-age groups, 15–19 and 20–24.

Because the malignant disease in younger patients tends to be more aggressive, thus possibly causing other malignancies, one might suspect that the younger patients are more prone to the multiple malignancies. The numbers of patients with other primary cancers $(Y_1, \ldots, Y_k)$ form a conditional multinomial random vector with parameters $(n = 11, k = 13, p_1, \ldots, p_k)$. If all the patients are equally susceptible, then p_i would be proportional to the expected number E_i in the ith age group. Letting $w_i = E_i/\sum E_j$, we write

$$p_i = w_i\pi_i \ ,$$

where π_i accounts for difference in susceptibility. The alternative hypothesis of a decreasing trend as age advances is expressed by

$$\pi_i > \pi_2 > \ldots > \pi_{13}, \ \pi_i/\pi_{i+1} \geq \lambda_i^* > 1 \ \text{for all } i.$$

With λ_i^*'s unknown, the maximin test statistic is

$$V(\tilde{a}^*, \tilde{y}) = \sum (14 - i)y_i \ .$$

The observed value of the statistic is 67 and the associated *exact* p value is .0044. (The exact p value is found from enumerating the exact null multinomial distribution.) The corresponding male-patient data in Bellet et al. (1977) yields the *exact* p value .0631.

5. DISCUSSION

This article shows that trend tests such as $\sum iy_i$ and $\sum x_iy_i$, which have appeared in the literature, may be justified as maximin tests against certain classes of ordered alternatives, and Theorem 1 also allows one to find such tests in nonstandard cases. Because the resulting tests are of the form $\sum a_iy_i > c$, a normal approximation to the multinomial usually yields sufficiently accurate p values. However, in doubtful cases, exact enumeration of multinomial probabilities or Monte Carlo estimates may be required.

APPENDIX

Lemma 1: Let $\tilde{\pi}$ and $\tilde{\pi}'$s be in $\Omega_k(\tilde{\lambda}^*)$ and satisfy $\pi_{l+1}/\pi_l \geq \lambda_l^*$ and $\pi'_{l+1}/\pi'_l = \lambda_l^*$ for some value of $l(1 \leq l \leq k-1)$ and $\pi_i = \pi'_i$ for $i \neq l$ or $l+1$. Then for each $\psi_a(\tilde{y}) \in \Psi_\alpha(A)$, the power β of ψ_a at $\tilde{\pi}'$ and $\tilde{\pi}$ satisfies

$$\beta_{\psi_a}(\tilde{\pi}') \leq \beta_{\psi_a}(\tilde{\pi}) \ .$$

Proof: The same as one for Lemma 1 of Lee (1977), except that $f(w_i, \pi_i)$ replaces p_i.

Lemma 2: Let $\tilde{\pi}_0 \in \Omega_k(\tilde{\lambda}^*)$ be an initial row vector. Let k be even. Define a sequence of row vectors $\{\tilde{\pi}_\nu\}_{\nu=1,\infty}$ by:

$$\pi_{\nu,1}\lambda_1^* = \pi_{\nu,2}$$

such that

$$f(w_1, \pi_{\nu-1,1}) + f(w_2, \pi_{\nu-1,2}) = f(w_1, \pi_{\nu,1}) + f(w_2, \pi_{\nu,2}) \ ,$$

$$\ldots,$$

$$\pi_{\nu,k-1}\lambda_{k-1}^* = \pi_{\nu,k}$$

such that

$$f(w_{k-1}, \pi_{\nu-1,k-1}) + f(w_k, \pi_{\nu-1,k}) = f(w_{k-1}, \pi_{\nu,k-1}) + f(w_k, \pi_{\nu,k})$$

for ν odd;

$$\pi_{\nu,1} = \pi_{\nu-1,1} ,$$
$$\pi_{\nu,2}\lambda_2^* = \pi_{\nu,3}$$

such that

$$f(w_2, \pi_{\nu-1,2}) + f(w_3, \pi_{\nu-1,3}) = f(w_2, \pi_{\nu,2}) + f(w_3, \pi_{\nu,3}) ,$$

$\ldots,$

$$\pi_{\nu,k-2}\lambda_{k-2}^* = \pi_{\nu,k-1}$$

such that

$$f(w_{k-2}, \pi_{\nu-1,k-2}) + f(w_{k-1}, \pi_{\nu-1,k-1}) = f(w_{k-2}, \pi_{\nu,k-2}) + f(w_{k-1}, \pi_{\nu,k-1}) ,$$

and

$$\pi_{\nu,k} = \pi_{\nu-1,k}$$

for ν even.

If k is odd, reverse "ν odd" and "ν even" properly. Then for any ν,

$$\tilde{\pi}\in\Omega_k(\tilde{\lambda}^*) \quad \text{and} \quad \lim_{\nu} \tilde{\pi}_\nu = \tilde{\pi}^* .$$

(See Theorem 1 for the definition of $\tilde{\pi}^*$.)

Proof: Obviously $\tilde{\pi}_0$ and $\tilde{\pi}_1$ are in $\Omega_k(\tilde{\lambda}^*)$. By induction on ν, $\tilde{\pi}_\nu \in \Omega_k(\tilde{\lambda}^*)$. Each $f(w_i, \pi_i)$ is bounded from above by one and strictly increasing in π_i. Thus π_i is bounded from above. Also, $\pi_{\nu,1} \geq \pi_{\nu-1,1}$ for each ν. Thus the limit of $\pi_{\nu,1}$ exists. Note that

$$\lim \pi_{\nu,1} = \lim \pi_{\nu_o,1} = \lim \pi_{\nu_o,2}/\lambda_1^* ,$$

where ν_o is odd. At each ν_o,

$$f(w_1, \pi_{\nu_o-1,1}) + f(w_2, \pi_{\nu_o-1,2}) = f(w_1, \pi_{\nu_o,1}) + f(w_2, \pi_{\nu_o,2}) .$$

Since in the limit of ν_o $f(w_1, \pi_{\nu_o-1,1}) = f(w_1, \pi_{\nu_o,1})$, we have

$$\lim f(w_2, \pi_{\nu_o-1,2}) = \lim f(w_2, \pi_{\nu_o,2}) .$$

The function $f(.,.)$ is continuous and strictly increasing in π. Hence

$$\lim \pi_{\nu_o,2} = \lim \pi_{\nu_o-1,2} = \lim \pi_{\nu_e,2} ,$$

where ν_e is even. Therefore, $\lim \pi_{\nu,2} = \lim \pi_{\nu,1}\lambda_1^*$. At each ν_e,

$$\pi_{\nu_e,2} = \pi_{\nu_e,3}/\lambda_2^*$$

and

$$f(w_2, \pi_{\nu_e-1,2}) + f(w_3, \pi_{\nu_e-1,3}) = f(w_2, \pi_{\nu_e,2}) + f(w_3, \pi_{\nu_e,3})$$

Since $\pi_{\nu,2}$ converges and

$$\lim f(w_2, \pi_{\nu_e-1,2}) = \lim f(w_2, \pi_{\nu_e,2}) ,$$
$$\lim f(w_3, \pi_{\nu_e-1,3}) = \lim f(w_3, \pi_{\nu_e,3}) .$$

Thus

$$\lim \pi_{\nu_e,3} = \lim \pi_{\nu_e-1,3} = \lim \pi_{\nu_o,3} = \lim \pi_{\nu,3} = \lim \pi_{\nu,2}\lambda_2^* = \lim \pi_{\nu,1}\lambda_1^*\lambda_2^* .$$

The lemma follows by repeating the arguments.

Lemma 3: Under the condition of Theorem 1,

$$\min_{\tilde{\pi}\in\Omega_k(\tilde{\lambda}^*)} E\{\psi_a(\tilde{Y})\,|\,\tilde{\pi}\} = E\{\psi_a(\tilde{Y})\,|\,\tilde{\pi}^*\}$$

for each $\psi_a(\tilde{y}) \in \Omega_\alpha(A)$.

Proof: The same as for one for Theorem 1 in Lee (1977) except that $f(w_i, \pi_i)$ replaces p_i.

[*Received December 1978. Revised March 1980.*]

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